

homework-2

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Question 1

```
install.packages("aRtsy")
```

The function “install.packages()” lets you download pre-made functions and datasets for use in future sessions. Once you install them, you won’t need to do it again.

The function “library()” loads the installed package in your current R session. It’s similar to checking the box next to the package you want to use in the Packages tab. You might want to dedicate a code chunk to all the packages you’ll use so your document renders properly.

Packages

Question 2

```
library(aRtsy)
```

Collatz Conjecture:

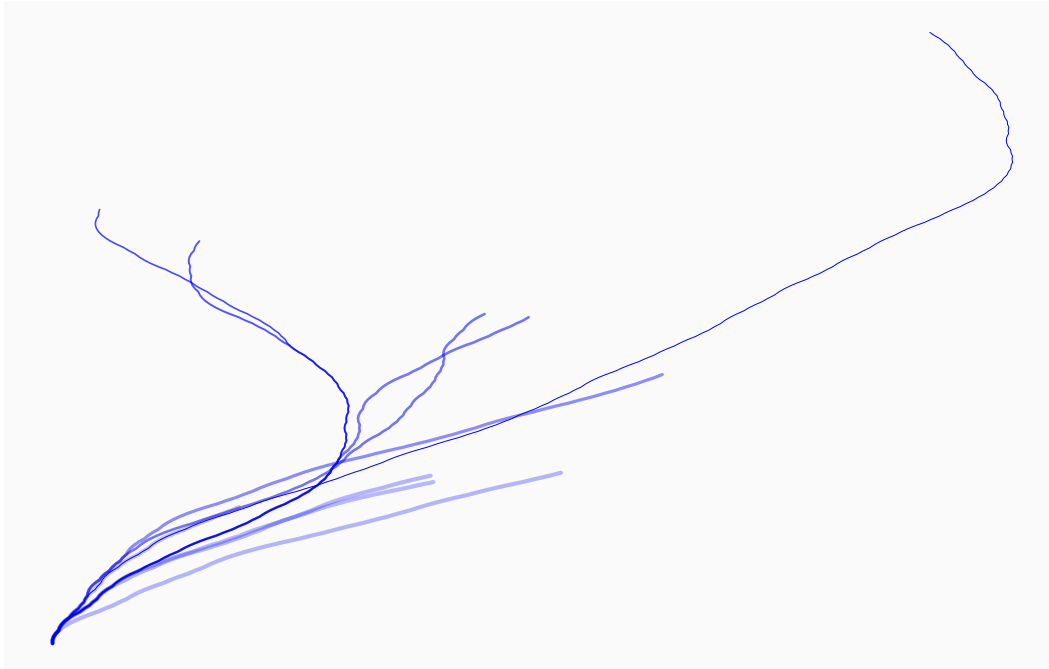
$$f(n) = \begin{cases} \frac{n}{2} & \text{if } n \equiv 0 \pmod{2} \\ \frac{3n+1}{2} & \text{if } n \equiv 1 \pmod{2} \end{cases}$$

The Collatz conjecture, or the $3n + 1$ problem, claims that no matter which positive integer you start with, you will eventually reach 1.

Question 3

a) Canvas_collatz()

```
canvas_collatz(  
  colors = c("blue"),  
  n = 10  
)
```



This ggplot shows how many steps (or the general path) it takes for a set of starting integers ($n = 10$) to reach 1. It's made to be plotted in blue.

b) Different Output?

- The function randomizes the 10 integers you start with. So, you could end up with a completely different plot each time you run it.
- This means it might look different and inconsistent each time you render it.
- With `set.seed()`, you use the same random set of integers associated with the seed number you provide, so you get the same result when you render it.

```
set.seed(9)

canvas_collatz(  
  colors = c("blue"),  
  n = 10  
)
```

